

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397465
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Aichele; Wilhelm

Rotary cutting device and method for operating a rotary cutting device

Abstract

A rotary cutting device is provided, which comprises a first and a second roller mounted rotatably on a machine stand. Either the first roller is a tool roller and the second roller is a counter-roller, or the second roller is a tool roller and the first roller is a counter-roller. The second roller is mounted on the machine stand to be displaceable on a first displacement axis. A cutting pressure device is provided by way of which a cutting pressure can be exerted between the second roller and the first roller. In a support mode, the second roller is supported against a wedge device. The wedge device has at least one displaceable wedge element. A position of the at least one displaceable wedge element predetermines a spacing between the second roller and the first roller. A drive device enables movement of the at least one displaceable wedge element.

Inventors:	Aichele; Wilhelm (Crailsheim, DE)
Applicant:	Aichele Werkzeuge GmbH (Crailsheim, DE)
Family ID:	1000008779454
Assignee:	Aichele Werkzeuge GmbH (Crailsheim, DE)
Appl. No.:	16/949885
Filed:	November 19, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210078194 A1	Mar. 18, 2021

Foreign Application Priority Data

DE	10 2018 112 310.8	May. 23, 2018
----	-------------------	---------------

Related U.S. Application Data

Publication Classification**Int. Cl.:** B26D7/26 (20060101)**U.S. Cl.:****CPC** B26D7/265 (20130101);**Field of Classification Search****CPC:** B26D (7/265)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5001950	12/1990	Fokos	83/37	B26D 7/265
2004/0003699	12/2003	Welch	N/A	N/A
2005/0229762	12/2004	Flaherty	83/508.3	B23D 35/007
2006/0257193	12/2005	Aichele	400/621	B26D 7/265
2008/0237386	12/2007	Aichele	N/A	N/A
2015/0090089	12/2014	Horii	83/344	B26D 7/265
2016/0279820	12/2015	Boutron	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1150620	12/1972	CA	N/A
2912458	12/1979	DE	N/A
29715037	12/1996	DE	N/A
102004050443	12/2005	DE	N/A
102005022604	12/2005	DE	N/A
102007016451	12/2007	DE	N/A
102013110510	12/2014	DE	N/A
1721712	12/2005	EP	N/A
2045144	12/1979	GB	N/A

Primary Examiner: Dong; Liang*Attorney, Agent or Firm:* Lipsitz & McAllister, LLC

Background/Summary

(1) This application is a continuation of international application number PCT/EP2019/063319 filed on 23 May 2019 and claims the benefit of German application number 10 2018 112 310.8 filed on 23 May 2018, which are incorporated herein by reference in their entirety and for all purposes.

BACKGROUND OF THE INVENTION

(1) The invention relates to a rotary cutting device, comprising a machine stand, a first roller that is mounted rotatably on the machine stand, a second roller that is mounted rotatably on the machine stand, wherein either (i) the first roller is a tool roller and the second roller is a counter-roller, or (ii) the second roller is a tool roller and the first roller is a counter-roller, and wherein the second roller is mounted on the machine stand such that it is displaceable on a first displacement axis, and a cutting pressure device by way of which a cutting pressure is configured to be exerted between the second roller and the first roller.

(2) Further, the invention relates to a method for operating a rotary cutting device, in which, during a machining operation, a first roller is supported against a second roller, and a material web is guided between the first roller and the second roller, wherein (i) the first roller is a tool roller and the second roller is a counter-roller, or (ii) the second roller is a tool roller and the first roller is a counter-roller, and wherein the second roller is displaceable in relation to the first roller on a first displacement axis.

(3) DE 297 15 037 U1 discloses a rotary punching machine or an impression apparatus, which has a roller and a counter-pressure cylinder that is arranged to form an intermediate space parallel to the roller and at a small spacing therefrom, wherein the roller and the counter-pressure cylinder are mounted rotatably and are coupled by means of a transmission for the purpose of simultaneous rotation, and where there can be introduced into the intermediate space a material web out of which workpieces can be fully or partly punched or pressed during rotation of the roller. Conical rolls in mutual abutment can be mounted on the axes for the roller and the counter-pressure cylinder. At least one of the axes is displaceable both axially and radially in relation to the other axis.

(4) DE 10 2004 050 443 A1 discloses a device for punching, having as the punching cylinder a first cylinder, which is rotatable about a punching cylinder axis of rotation, and having as the counter-cylinder a second cylinder, which is rotatable about a counter-cylinder axis of rotation that runs axially parallel to the punching cylinder axis of rotation, wherein there is configured to be formed between the punching cylinder and the counter-cylinder a punch gap size that is adjustable by means of at least one adjustment device.

(5) DE 29 12 458 A1 discloses a rotary punch for punching envelope blanks out of moving material webs of paper or similar, or for punching shaped cuts out of premanufactured envelope blanks using a counter-roller supported against the bladed roller, having single thrust bearings between which there is located an adjustable body.

(6) DE 10 2013 110 510 A1 discloses a device for rotary punching, having a punching cylinder that is rotatable about a punching cylinder axis, having a counter-pressure cylinder that is rotatable about a counter-pressure cylinder axis, wherein the counter-pressure cylinder has raceways on which the punching cylinder or its punching cylinder raceways can run by way of running surfaces, and having an adjustment device by which a gap size between the punching cylinder and the counter-pressure cylinder is adjustable, and having a further cylinder that takes the form of a support shaft on which the counter-pressure cylinder is directly supported.

(7) DE 10 2007 016 451 A1 discloses a rotary cutting device, comprising a machine stand, a cutting roller that is mounted on the machine stand, and a counter-roller that is mounted on the machine stand, wherein the cutting roller and/or the counter-roller have an inner core and an outer sheath that is arranged around the inner core.

(8) DE 10 2005 022 604 A1 discloses a rotary cutting device having a rotatably mounted cutting roller and a counter-roller, wherein at least one support ring is provided for the purpose of supporting the cutting roller against the counter-roller. A raising device is provided for the purpose of moving the cutting roller and the counter-roller apart.

SUMMARY OF THE INVENTION

(9) In accordance with an embodiment of the invention, a rotary cutting device is provided that has

a high degree of operational reliability.

(10) In accordance with an embodiment of the invention, provision is made in the rotary cutting device that, in a support mode, the second roller is supported against a wedge device, that the wedge device has at least one displaceable wedge element, wherein a displacement position of the at least one displaceable wedge element predetermines a spacing between the second roller and the first roller, and that there is associated with the at least one displaceable wedge element a drive device for a displacement movement of the at least one displaceable wedge element on a second displacement axis.

(11) During a machining operation (such as a cutting operation) of the rotary cutting device, a material web is guided between the first roller and the second roller, wherein in particular the first roller and the second roller rotate at the same speed of revolution. A corresponding cutting force is set by way of the cutting pressure device. In this context, in particular the second roller is pressed against the first roller with the required force. Force is correspondingly introduced by way of a roller bearing.

(12) In principle, the spacing between the second roller and the first roller during a machining operation for a workpiece (such as a cutting operation) is fixed, and in particular the second roller is supported against the first roller.

(13) It may arise that the material web includes “disruptions”. Such disruptions may be foreign bodies such as screws, tools, etc. that have found their way onto the material web. For example, there may also be deformed products, doubled products, etc. on the material web, or the material web may be skewed.

(14) As a result of disruptions of this kind, the material web may be thicker in some regions than the normal thickness. This in turn can result, if the second roller is displaceable, in the second roller being raised away, in which case, because of the introduction of force by way of the cutting pressure device, in principle the second roller may bounce back onto the first roller.

(15) In the event of bouncing back in this way, there may be damage to the tool roller, with damage to a blade, and/or to the counter-roller, for example resulting in scoring.

(16) According to the invention, protection against roller impact is provided. In the support mode, in particular in addition to a direct support against the first roller, the second roller is also supported by way of the wedge device.

(17) In particular in this case, the second roller is additionally supported on the machine stand by way of the wedge device. If the first roller is fixed in a manner preventing displacement relative to the machine stand, there is an additional indirect support of the second roller against the first roller by way of the wedge device.

(18) If the second roller is raised away from the first roller as a result of a disruption in the material web, then the drive device ensures a displacement of the at least one displaceable wedge element on the second axis of displacement in order to establish support again, but in this case the support prevents the second roller from impacting against the first roller, because displacement of the at least one wedge element allows the spacing between the second roller and the first roller to be increased.

(19) In this way, effective protection against roller impact is provided in a structurally simple manner.

(20) The wedge device can be integrated in the rotary cutting device and in particular in the machine stand in a manner taking up relatively little space.

(21) The drive device for the displacement movement of the at least one displaceable wedge element may take a structurally simple form. For example, the drive device may take the form of a pneumatic or hydraulic cylinder or a motor (electric motor). The displacement movement may be performed in a structurally simple embodiment by means of a biased drive device, automatically and without the complexity of any open or closed-loop control.

(22) It is most particularly advantageous if, during a machining operation (such as a cutting

operation), the first roller is supported against the second roller by way of at least one support ring, which is arranged on the first roller and/or the second roller. As a result of the at least one support ring, during a machining operation contact, or “excessive contact”, between a blade of the tool roller and the counter-roller is prevented. It is possible to make a precise cut, and it is possible to set an optimised cutting force.

(23) In one exemplary embodiment, a first support ring and a second support ring are arranged on a tool roller, wherein in particular a blade is positioned between the first support ring and the second support ring in relation to a direction parallel to an axis of rotation of the tool roller. This enables an effective machining operation (such as a cutting operation) to be achieved.

(24) It is most particularly advantageous if the drive device takes a form such that a movement of the at least one displaceable wedge element is driven in a direction in which a spacing between the second roller and the first roller is increased if the second roller is supported against the wedge device. This enables a protection against roller impact to be achieved. As a result of the displaceable wedge element correspondingly following by way of the drive device, a re-established support of the second roller against the wedge device prevents impact against the first roller.

(25) It may be provided for the drive device to take a form such that it creates a bias acting on the at least one displaceable wedge element. In particular, during a machining operation a constant bias applies. In this arrangement, during the machining operation a counter-force exerted by way of the cutting pressure device prevents the at least one displaceable wedge element from being displaced. If the second roller is raised away from the first roller after meeting a disruption, and as a result this counter-force is reduced, then the bias of the drive device has the effect of automatically displacing the at least one displaceable wedge element, and support of the second roller against the wedge device is re-established, but in this case the spacing between the second roller and the first roller is increased and hence in turn a protection against roller impact is provided—that is to say that the second roller cannot impact against the first roller.

(26) In principle, it is possible, if a bias acting on the at least one displaceable wedge element is provided (in the support mode), for this bias not to be subject to closed-loop control. In that case, the bias is predetermined by the drive device. In one embodiment, closed-loop control of the bias is provided. The bias is adapted to the prevailing conditions at the rotary cutting device. For example, the position of a wedge element in the support mode may be changed as a result of guidance play and vibrations in the rotary cutting device. As a result of closed-loop control of the bias, such changes may be taken into account and in particular their effect on the corresponding wedge element may be compensated. In one embodiment, for this purpose there is provided a sensor device that detects movement and/or a change in position of the at least one displaceable wedge element and communicates this to an evaluation device. The evaluation device controls the drive device accordingly in order to adjust the bias in dependence on a detected movement or position of the corresponding wedge element and hence to effect closed-loop control of the bias in a feedback loop.

(27) In an alternative embodiment, the drive device takes a form such that it does not create any bias acting on the at least one displaceable wedge element. As a result, in particular there is no need for compensation of a displacement or change in position of the at least one displaceable wedge element in the support mode. Movements or changes in position of this kind may arise for example as a result of guidance play and/or vibrations in the rotary cutting device. If a bias is provided, movements or changes in position of this kind may be reflected in the bias. If the drive device takes a form such that it can perform its function even in the absence of bias, then changes in movement of this kind do not have a negative effect in the support mode, and need not be compensated.

(28) It is favorable if there is provided a sensor device that detects a position of the second roller in relation to the machine stand, wherein in particular the sensor device takes the form of a distance-sensor device or position-sensor device. As a result of the sensor device, in particular the fact that the second roller has been raised away is detected, and hence termination of the support mode can

be detected. This in turn can be used to control the drive device such that a corresponding displacement movement of the at least one displaceable wedge element is performed. In particular, the sensor device is advantageous if no bias is provided between the drive device and the wedge device.

(29) It is favorable if the sensor device is connected, in a manner configured to transfer signals, to an evaluation device, and the evaluation device controls the drive device in dependence on signals from the sensor device. By way of the sensor device, the evaluation device may detect a position, and in particular a raised-away position, of the second roller. This may then be used for a corresponding control of the drive device and movement of the at least one displaceable wedge element.

(30) It is most particularly advantageous if the evaluation device controls the drive device such that, when the sensor device detects a threshold value that corresponds to the second roller being raised away from the first roller and to the second roller no longer being supported against the wedge device, the at least one displaceable wedge element is displaced and in particular is automatically displaced such that the second roller is once again supported against the wedge device, in which case the second roller is at a spacing from the first roller. In this way, it is possible to protect against roller impact automatically, using the detection results of the sensor device. This protection against roller impact is automated. It can be achieved independently of movements or changes in position of the at least one displaceable wedge element in the support mode (caused for example by guidance play and/or vibrations).

(31) It is further favorable if the drive device takes a form such that, when the second roller is raised relatively away from the first roller and the second roller is no longer supported against the wedge device, the at least one displaceable wedge element is displaced automatically such that the second roller is once again supported against the wedge device, in which case the second roller is at a spacing from the first roller. This achieves effective protection against roller impact in a structurally simple manner. In particular, this protection against roller impact can be achieved without the complexity of any open or closed-loop control.

(32) In particular, the support mode, in which the second roller is supported against the wedge device and as a result in particular on the machine stand, prevails during a normal machining operation, and prevails if, after the second roller has been raised relatively away from the first roller, the at least one displaceable wedge element has been displaced for the purpose of the second roller being once again supported against the wedge device. During the machining operation, in particular the second roller is supported against the first roller by way of at least one support ring, and in addition there is support by way of the wedge device. The support mode is briefly canceled if there is in the material web a corresponding disruptive body that results in the second roller being raised away from the first roller. During this raising-away action, the second roller is not supported against the wedge device. The support mode prevails again only once the at least one movable wedge element has followed such that support is once again provided. There is then no longer any (additional) support by way of a support ring.

(33) In one embodiment, the wedge device comprises at least one part-device having a first wedge element and a second wedge element, wherein in the support mode the first wedge element is supported against the second wedge element, and wherein the first wedge element and/or the second wedge element is displaceable and coupled to the drive device. In this way, in the support mode it is possible to achieve permanent support of the second roller against the wedge device. Further, the at least one displaceable wedge element can be made to track in a simple manner in order, if a support mode is briefly canceled, to achieve the support mode again.

(34) In particular in this case, the first wedge element is associated with the first roller and the second wedge element is associated with the second roller.

(35) In one embodiment, the first wedge element is connected to the first roller in a manner preventing displacement relative to the first axis of displacement. If the first roller is prevented

from being displaced on the first axis of displacement relative to the machine stand, then the first wedge element is also arranged on the machine stand in a manner preventing displacement relative to the first axis of displacement. In that case it is advantageous from a structural point of view if the first wedge element is displaceable, since displaceability on the first axis of displacement is achievable in a manner preventing displacement relative to the machine stand. In particular, it is then possible to guide the first wedge element on a displacement guide that is prevented from displacement relative to the machine stand.

(36) It may further be provided for the second wedge element to be connected to the second roller in a manner preventing displacement relative to the first axis of displacement. In that case, if the second roller is displaceable relative to the machine stand on the first axis of displacement, it is provided in particular for the second wedge element to be displaceable (relative to the machine stand) with the second roller on the first axis of displacement. This makes it possible in particular in an effective and structurally simple manner to achieve protection against roller impact once the second roller has been raised away from the first roller.

(37) In one embodiment, the first wedge element is arranged on a first bearing housing of the first roller, by way of which the first roller is seated on the machine stand. The first roller is mounted by way of the bearing housing such that it is rotatable about an axis of rotation.

(38) It is further favorable if the second wedge element is arranged on a second bearing housing by way of which the second roller is seated on the machine stand. The second roller is mounted by way of the bearing housing such that it is rotatable. In particular, the second roller is displaceable relative to the machine stand by way of the bearing housing, also on the first axis of displacement.

(39) In principle, it is possible for both the first roller and the second roller to be displaceable relative to the machine stand (on the first axis of displacement). In a structurally simple embodiment, the first roller is positioned on the machine stand in a manner preventing displacement in relation to the first axis of displacement. For example, in that case the second roller, which is seated on the machine stand such that it is displaceable in relation to the first roller on the first axis of displacement, is seated above the first roller in relation to the direction of gravity.

(40) It is favorable if the first axis of displacement is oriented transversely and in particular perpendicular to the second axis of displacement. This produces an effective protection against roller impact, with a structurally simple construction of the protection against roller impact. The corresponding wedge device can be integrated into the rotary cutting device in a manner taking up little space. For example, the first axis of displacement is an axis parallel to the direction of gravity. The second axis of displacement is in that case in particular a horizontal axis.

(41) It is further favorable if the first axis of displacement and/or the second axis of displacement are oriented transversely and in particular perpendicular to an axis of rotation of the first roller. This produces an effective cutting operation with a way of protecting against roller impact that is of structurally simple construction.

(42) In one embodiment, the wedge device has a first part-device and a second part-device, wherein the first part-device and the second part-device are at a spacing from one another in a direction parallel to an axis of rotation of the first roller, and by means of the support mode the second roller is supported both against the first part-device and against the second part-device. This produces a symmetrical support of the second roller in relation to the machine stand.

(43) In particular, the first part-device and the second part-device take the same form, producing an effective support and also an effective protection against roller impact. In particular, once the second roller has been raised away from the first roller, the mutual orientation of the axes of rotation of the first roller and the second roller is not changed, and in particular they remain in a parallel orientation.

(44) It is further favorable if the drive device has a first drive for the first part-device and a second drive for the second part-device, wherein in particular the first drive and the second drive are

synchronized. In this way, an effective protection against roller impact can be achieved. The orientation of the axes of rotation of the first roller and the second roller is maintained, and in particular the parallel arrangement of the axes of rotation is maintained.

(45) In one embodiment, the drive device takes a form such that it constantly exerts a force on the at least one displaceable wedge element. In that case, it is in particular provided, during a machining operation and taking into account the force exerted by way of the cutting pressure device, for the force exerted by the drive device to be insufficient to displace the at least one displacement element. Only once there is a reduction in force, arising from the fact that the second roller is raised away from the first roller, does the constant application of force result in a displacement of the at least one movable wedge element. In this way, in a simple manner, the drive device can be used to achieve a type of bias at the wedge device. The at least one movable wedge element can be displaced in a simple manner in the event of a malfunction with no need for example for a complex closed-loop control circuit. It is also possible for a force subject to closed-loop control (a bias subject to closed-loop control) to be constantly exerted on the at least one displaceable wedge element. It is furthermore possible for there to be no permanent bias acting on the displaceable wedge element.

(46) The drive device is or comprises for example a mechanical drive (in particular having a spring device) or pneumatic drive (having one or more pneumatic cylinders) or hydraulic drive (having one or more hydraulic cylinders) or magnetic drive or inductive drive or electromagnetic drive or a motorized drive. The drive device may take a relatively compact form and can be integrated into the rotary cutting device in a manner taking up little space.

(47) In particular, the tool roller is a cutting roller or stamping roller or bladed roller or compactor roller or squeezing roller. The tool roller performs workpiece machining, in particular in a defined region of the workpiece. For example, in the case of a cutting roller the shape of a blade predetermines the region of the workpiece that is to be machined.

(48) In one embodiment, the first roller and/or the second roller are supported by at least one further roller. This makes it possible to prevent excessive deflection of the first roller or the second roller.

(49) It is in particular provided, during a machining operation, for displaceability of the at least one displaceable wedge element to be released and for the drive device to act on the at least one displaceable wedge element, and for a cutting force to be set between the first roller and the second roller by way of the cutting pressure device, wherein the wedge device and the drive device take a form adapted to one another such that during disruption-free operation the second roller is pressed against the first roller (and in particular the second roller is also supported against the first roller by way of at least one support ring), and the spacing between the first roller and the second roller is fixed. In particular, this makes it possible during the cutting operation for the first roller to be supported directly against the second roller (in particular by way of support rings). In addition, there is an in particular indirect support by way of the wedge device. This provides a starting position for achieving effective protection against roller impact.

(50) In that case it is provided, when the second roller is raised away from the first roller because of a disruption in a material web, for the drive device to displace the at least one displaceable wedge element (preferably automatically) such that it is not possible, because support of the second roller against the wedge device is then re-established, for the second roller to impact against the first roller. Taking as a starting point a support mode, when the second roller is raised away from the first roller there is briefly no longer any support against the wedge device. After the at least one displaceable wedge element has been pushed in, support is re-established (wherein in particular there is no longer support by a support ring). However, in the condition with this re-established support, the at least one displaceable wedge element is displaced in relation to its starting position in the cutting operation, and so the spacing between the second roller and the first roller is greater than in the machining operation. This effectively prevents impact of the second roller against the

first roller. Conventionally, the rotary cutting device then has to be switched off, which can take place in particular automatically. A machining operation is only possible again after the wedge device having the at least one displaceable wedge element is put into its starting position for the machining operation.

(51) It is advantageous if there is provided a sensor device that determines a displacement position of the at least one displaceable wedge element and in particular determines it in relation to the machine stand. It is then possible, using the sensor device, to detect that the second roller has been raised away from the first roller. This is manifested by a change in the displacement position of the at least one displaceable wedge element. The result is a disruption in normal operation, and then, in particular as a result of the corresponding sensor signals, the rotary cutting device is switched off.

(52) It is then advantageous if there is provided a display device and/or an evaluation device that is coupled to the sensor device in a manner configured to transfer signals. If a new displacement position of the at least one displaceable wedge element that must have been caused by a disruption is detected, then by way of the evaluation device the rotary cutting device can be switched off in order to prevent further possible damage. Further, it is then also possible to detect product-related disruptions at the material web, and corresponding disruptions can be eliminated. A corresponding disruption can be displayed by way of the evaluation device.

(53) According to the invention, a method of the type mentioned in the introduction is provided in which, in a support mode, the second roller is supported against a wedge device, and at least one movable wedge element is displaced such that, when the second roller is raised away from the first roller, the second roller is once again supported against the wedge device, wherein the second roller is then positioned at a spacing from the first roller.

(54) The method according to the invention has the advantages that have already been explained in connection with the rotary cutting device according to the invention.

(55) In particular, the rotary cutting device according to the invention can be operated using the method according to the invention, or the method according to the invention can be performed on the rotary cutting device according to the invention.

(56) In this case, it is advantageous if the displacement movement of the at least one displaceable wedge element is performed automatically. As a result, there is no need for the complexity of any open-loop control or closed-loop control. This produces a high degree of operational reliability.

(57) Favorably, the at least one displaceable wedge element is displaced automatically such that, once the second roller has been raised away from the first roller, the possibility that the second roller will impact against the first roller is prevented because the second roller is supported against the wedge device. The method according to the invention can be performed without the complexity of any open-loop control or closed-loop control. The corresponding wedge device can be integrated into the corresponding rotary cutting device in a manner taking up little space.

(58) It is favorable that, during the machining operation, the first roller and the second roller are operated at the same speed of revolution. This produces a high cutting performance with effective throughput.

(59) The description below of preferred embodiments serves, in connection with the drawings, to explain the invention in more detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an isometric representation of an exemplary embodiment of a rotary cutting machine according to the invention;

(2) FIG. 2 shows a partial representation of the region A of the rotary cutting machine according to FIG. 1, with a sensor device;

- (3) FIG. 3(a) shows a side view of the rotary cutting machine according to FIG. 1, in a non-active mode;
- (4) FIG. 3(b) shows a similar view to FIG. 3(a), during a cutting operation;
- (5) FIG. 3(c) shows a front view of the rotary cutting machine according to FIG. 1, during the cutting operation (FIG. 3(c) corresponds to FIG. 3(b));
- (6) FIG. 4(a) shows the same view as FIG. 3(a) after a disruption, with a wedge element displaced;
- (7) FIG. 4(b) shows a front view of the rotary cutting machine according to FIG. 1, in the case of disruption according to FIG. 4(a);
- (8) FIG. 5 shows an isometric representation of a further exemplary embodiment of a rotary cutting device according to the invention;
- (9) FIG. 6 shows a variant of the rotary cutting machine according to FIG. 1, in a front view; and
- (10) FIG. 7 shows a variant of the rotary cutting device according to FIG. 5, in a front view.

DETAILED DESCRIPTION OF THE INVENTION

- (11) An exemplary embodiment of a rotary cutting device **10** according to the invention (FIGS. 1 to 4(b)) comprises a machine stand **12**. In one embodiment, the machine stand **12** has a base **14** by way of which the rotary cutting device **10** is mountable on a substructure.
- (12) Arranged on the base **14** is a frame **16**. The frame **16** comprises bearers **18** oriented transversely in relation to the base **14** and fixed thereto.
- (13) When the rotary cutting device **10** is mounted in the intended manner, the bearers **18** are oriented in particular parallel to the direction of gravity *g*.
- (14) Neighboring bearers **18** are connected to one another by way of transverse struts **20**.
- (15) In particular, the frame **16** comprises a first frame element **22**, which is composed of two bearers at a spacing, and a transverse strut arranged in between. Further, the frame **16** comprises a second frame element **24** that takes at least approximately the same form as the first frame element **22** and is seated on the base **14**, at a spacing from the first frame element **22**. The first frame element **22** and the second frame element **24** are connected to one another by way of one or more struts **26**, wherein the strut or struts **26** are seated in an upper region of the respective frame elements **22**, **24** and at a spacing from the base **14**.
- (16) A first roller **30** is mounted on the frame **16** by way of a first bearing housing **28** such that it is rotatable about an axis of rotation **32**. Here, the first roller **30** is positioned between the first frame element **22** and the second frame element **24**. The first bearing housing **28** is fixed to the first frame element **22** and the second frame element **24** respectively.
- (17) In one embodiment, the first roller **30** is held on the frame **16** in a manner preventing displacement.
- (18) In the exemplary embodiment shown, the first roller **30** is a tool roller **34** such as a cutting roller. This has a first support ring **36** and, at a spacing, a second support ring **38**. Here, the first support ring **36** and the second support ring **38** are spaced from one another in a direction parallel to the axis of rotation **32**.
- (19) Seated between the first support ring **36** and the second support ring **38** on the tool roller **34** is a blade **40** that takes a form such that a corresponding region of the desired shape is configured to be cut out of a material web.
- (20) Further seated on the frame **16** is a second roller **42**. The second roller **42** is mounted on a second bearing housing **44** such that it is rotatable about an axis of rotation **46**. The axis of rotation **46** is parallel to the axis of rotation **32** of the first roller **30**.
- (21) The second bearing housing **44** is fixed to the frame **16**. The second roller **42** is positioned such that it is oriented with the first roller **30** between the first frame element **22** and the second frame element **24**.
- (22) If the first roller **30** is a tool roller, than the second roller **42** is a counter-roller (anvil roller) **48**.
- (23) During a cutting operation of the rotary cutting device **10**, the counter-roller **48** is supported against the tool roller **34** by way of the support rings **36**, **38**. A material web can pass between the

support rings **36, 38**, against the counter-roller **48** and the tool roller **34**.

(24) It is provided in particular for the counter-roller **48** to have a length, in a direction parallel to the axis of rotation **32** or **46**, that is greater than the length of the tool roller **34** in the same direction between the first support ring **36** and the second support ring **38**.

(25) It is provided in particular for a rotary drive for the rotary movement of the first roller **30** about the axis of rotation **32** to be seated on the first bearing housing **28**. In particular, a drive for a rotary movement of the second roller **42** about the axis of rotation **46** is seated on the second bearing housing **44**.

(26) For a cutting operation of the rotary cutting device **10**, it is provided in particular for the first roller **30** and the second roller **42** to be driven at the same speed of revolution.

(27) In the embodiment shown, the second roller **42** is seated above the first roller **30**, in relation to the direction of gravity g .

(28) In principle, it is also possible for the tool roller **34** to be seated above the counter-roller **48**.

(29) The second roller **42** is mounted on the machine stand **12** such that it is displaceable on the machine stand **12** on a first axis of displacement **50**.

(30) The first axis of displacement **50** lies transversely and in particular perpendicular to the axis of rotation **32** or **46**.

(31) In particular, the first axis of displacement **50** is parallel to the direction of gravity g if the machine stand **12** is mounted on a substructure by way of its base **14**.

(32) For the purpose of guiding displacement of the second roller **42** on the machine stand **12**, there is provided a corresponding guide device.

(33) Seated on the machine stand **12** is a cutting pressure device **52** by which a cutting force is settable between the tool roller **34** and the counter-roller **48** during a cutting procedure. The cutting pressure device **52** presses the second roller **42** against the first roller **30** (supported at the support rings **36, 38**) with the desired force. As a result of the displaceability of the second roller **42** on the first axis of displacement **50**, this accordingly allows a cutting force or a cutting pressure to be set at a material web.

(34) The cutting pressure device **52** has in particular corresponding setting arrangements, such as one or more hydraulic cylinders or one or more pneumatic cylinders. In principle, the cutting pressure device **52** may also have motorized drives for setting the cutting force.

(35) In one exemplary embodiment, the cutting pressure device **52** has a first sub-unit **54** and a second sub-unit **56**. These allow the second roller **42** (the counter-roller **48**) to be pressed with the desired force, on the first axis of displacement **50**, in the direction of the first roller **30** (the tool roller **34**), with the result that the desired cutting force is set.

(36) In particular, by way of the cutting pressure device **52** a pressing force of the second roller **42** in a direction **58** toward the first roller **30** is set, on the first axis of displacement **50**.

(37) The second roller **42** is configured to be raised away from the first roller **30** in an opposing direction **60** to the direction **58**.

(38) A wedge device **62** is provided, by way of which the second roller **42** is supported in relation to the first roller **30** in a support mode (this is explained in more detail below).

(39) In the exemplary embodiment shown, the first roller **30** is arranged on the machine stand **12** in a manner preventing displacement (in particular on the first axis of displacement **50**). By way of the wedge device **62**, this allows the second roller **42** to be supported in relation to the machine stand **12** in the support mode.

(40) During a cutting operation, the counter-roller **48** is supported against the first roller **30** by way of the support rings **36, 38**. Further, it is supported on the machine stand **12** by way of the wedge device **62**.

(41) The wedge device **62** is arranged and formed correspondingly such that, during the cutting operation, support of the second roller **42** is possible both directly against the first roller **30**, by way of the support rings **36, 38**, and also indirectly, by way of the machine stand **12**.

(42) In particular, the wedge device **62** comprises a first part-device **64** that is associated with the first frame element **22**, and a second part-device **66** that is associated with the second frame element **24**. The first part-device **64** and the second part-device **66** take basically the same form.

(43) The first part-device **64** and the second part-device **66** are spaced from one another in a direction parallel to the axis of rotation **32** or **46**. Between them in relation to this direction there are positioned the first roller **30** and the second roller **42** respectively.

(44) The first part-device **64** and the second part-device **66** each have a first wedge element **68** and a second wedge element **70**. The first wedge element **68** is connected to the first bearing housing **28** in a manner preventing displacement in relation to the first axis of displacement **50**.

(45) The first wedge element **68** is displaceable in relation to the machine stand **12** or the first bearing housing **28** on a second axis of displacement **72**.

(46) The second axis of displacement **72** is transverse and in particular perpendicular in relation to the first axis of displacement **50**. Further, the second axis of displacement **72** is transverse and in particular perpendicular to the axis of rotation **32** or **46**.

(47) The first wedge element **68** of the wedge device **62** is guided on a corresponding displacement guide.

(48) In one embodiment, the second wedge element **70** is arranged on the second bearing housing **44** and is fixedly connected thereto. It is prevented from displacement in relation to the second bearing housing **44** both on the first axis of displacement **50** and on the second axis of displacement **72**.

(49) In the event of displacement of the second roller **42** (of the second bearing housing **44**) on the first axis of displacement **50**, the second wedge element **70** is displaced in synchronism therewith.

(50) The second wedge element **70** is adapted to the first wedge element **68**.

(51) The first wedge element **68** has a first support face **74**. The second wedge element **70** has a second support face **76**, opposed to the first support face.

(52) In a support mode, the second support face **76** lies on the first support face **74**.

(53) Both the first support face **74** and the second support face **76** are oblique faces. They have the same value in respect of inclination.

(54) In a support mode, a spacing between a lower side **78** of the first wedge element **68** and an upper side **80** of the second wedge element **70** is the same along the second axis of displacement **72**.

(55) A spacing between the lower side **78** of the first wedge element **68** and the first support face **74** varies along the second axis of displacement **72**.

(56) Further, a spacing between the upper side **80** of the second wedge element **70** and the second support face **76** varies along the second axis of displacement **72**. Here, the variation is in each case linear.

(57) In the exemplary embodiment shown, a height of the first wedge element **68** between the lower side **78** and the first support face **74** decreases in a direction **82**. A height of the second wedge element **70** between the upper side **18** and the second support face **76** decreases in a direction **84** that is an opposing direction to the direction **82**.

(58) Associated with the wedge device is a drive device, designated **86** as a whole. Here, the drive device **86** comprises a first drive **88**, which is associated with the first part-device **64**, and a second drive **90**, which is associated with the second part-device **66**.

(59) In this arrangement, the first drive **88** and the second drive **90** respectively act on the respective first wedge element **68** of the corresponding first part-device **64** and the second part-device **66**. By way of the respective drive **88** and **90**, the corresponding first wedge element **68** is displaced in the direction **82**.

(60) Here, the drive device **86** takes a form such that the first drive **88** and the second drive **90** are synchronized, with the result that the respective first wedge elements **68** of the first part-device **64** and the second part-device **66** are displaced in synchronism in the direction **82** (see below) and as a

result a parallel orientation of the axes of rotation **32** and **46** is maintained.

(61) In one embodiment, the drive device **86** takes a form such that it is biased. During active operation (cutting operation) of the rotary cutting device **10**, in particular the drive device **86** exerts a constant force on the respective wedge element **68**. In an alternative embodiment, described below with reference to FIGS. **6** and **7**, the drive device is formed without bias.

(62) During normal operation, the force is such that the corresponding first wedge element **68** does not undergo any movement.

(63) A counter-force of the second roller **42** acting on the wedge device **62**, and by way of the second wedge element **70** on the first wedge element **68**, is generated in particular by the cutting pressure device **52** and prevents the respective first wedge element **68** from being displaced.

(64) The drive device **86** having the first drive **88** comprises in particular a hydraulic drive or a pneumatic drive or a mechanical drive such as a spring device. In principle, it is also possible for the drive device **86** to comprise a motorized drive or an electric, electric motor or magnetic drive, etc.

(65) In the exemplary embodiment shown, at the first drive **88** and the second drive **90** respectively, the drive device **86** has a pneumatic cylinder **92** having connections **94**. This pneumatic cylinder **92** is coupled to the first wedge element **68**, or the second wedge element **70** in the case of the second drive **90**, and constantly urges it at a corresponding force.

(66) In this arrangement, it is possible for the bias created by the drive device **86** on the first wedge element **68** not to be, or to be, subject to closed-loop control. In the case of a bias not subject to closed-loop control, the bias is predetermined by the drive device **86**. For example, a constant force is applied. In the case of a bias subject to closed-loop control, the bias is adapted to the actually prevailing conditions. For example, this allows movements of the wedge element **68** resulting from guidance play and vibrations to be compensated.

(67) In the case of such an embodiment of a bias subject to closed-loop control, in particular a sensor device **91** is provided that detects a movement or a change in position in particular of the respective first wedge element **68**. Corresponding sensor signals are forwarded to an evaluation device **98** (see below). The evaluation device **98** then controls the first drive **88** and the second drive **90** respectively as regards the bias, and adapts this to the current state of the respective wedge element **68**.

(68) In particular, there is provided for the first wedge element **68** a respective separate sensor device **91**, associated with the first drive **88**, and there is provided another sensor device **91** for the corresponding wedge element associated with the second drive **90**.

(69) The fact that the sensor device **91** is connected to the evaluation device **98** in a manner configured to transfer signals is indicated schematically in FIG. **2** by a line having the reference numeral **91a**. The fact that the evaluation device **98** is connected to the first drive **88** in a manner configured to transfer signals, for the purpose of setting the bias at the first drive **88** in relation to the first wedge element **68**, is indicated schematically in FIG. **2** by a signal line having the reference numeral **91b**.

(70) Closed-loop control of the bias allows changes in the position of the respective wedge element **68**, for example as a result of guidance play and vibrations in the rotary cutting device **10**, to be detected and adjusted for or compensated. This closed-loop control of the bias is performed in this case in a support mode (see below), in which the second wedge element **70** is supported against the first wedge element **68** and hence the second roller **42** is indirectly supported against the first roller **30**.

(71) It is also possible—as explained in more detail below—for the drive device not to have any bias relative to the wedge device **62** (in the support mode).

(72) A decrease in the counter-force brings about a movement of the first wedge element **68** in the direction **82**.

(73) In one embodiment, there is associated with the wedge device **62** a sensor device **96**, by which

a displacement position of the respective first wedge element **68** or of only one wedge element **68** on its displacement guide is detectable, in particular in relation to the first bearing housing **28**.

(74) The sensor device **96** is connected to an evaluation device **98** and/or a display device in a manner configured to transfer signals. This makes it possible to determine whether the first wedge element **68** has been displaced.

(75) In particular, a displacement of the first wedge element **68** can, by way of the evaluation device **98**, result in a cutting operation of the rotary cutting device **10** being switched off.

(76) The rotary cutting device **10** operates as follows:

(77) When operation of the rotary cutting device **10** is inactive (FIG. 3(a)), the first wedge element **68** is positioned such that it is not possible to perform a cutting operation.

(78) During an active cutting operation of the rotary cutting device **10** (FIGS. 3(b) and (c)), a material web **100** (FIG. 3(c)) is guided between the counter-roller **48** and the tool roller **34**, and in so doing between the support rings **36**, **38**.

(79) During the cutting operation, the tool roller **34** is supported against the counter-roller **48** by way of its support rings **36**, **38**. Here, the counter-roller **48** is pressed against the tool roller **34** on the first displacement axis **50** with the desired force by way of the cutting pressure device **52** in order to set the corresponding cutting force.

(80) At the wedge device, the first wedge element **68** is positioned such that a support mode prevails and the second wedge element **70** is supported against the first wedge element **68**. This in turn causes the second roller **42** (the counter-roller **48**) to be indirectly supported against the first roller, and at the same time to be directly supported on the machine stand **12**.

(81) The counter-roller **48** is supported directly against the tool roller **34** by way of the support rings **36**, **38**.

(82) The position of the first wedge element **68** in the support mode and thereby during the cutting operation is such that a cutting operation is in fact performed on the material web **100** with the desired cutting force.

(83) It may occur that the material web **100** contains foreign bodies and in particular metal foreign bodies, such as screws, overlooked tools, and similar.

(84) If such foreign bodies come between the tool roller **34** and the counter-roller **48**, this causes the first roller **30** and the second roller **42** to be raised away from one another.

(85) In one embodiment, in which the second roller **42** (the counter-roller **48**) is mounted on the machine stand **12** displaceably on the first displacement axis **50**, such a foreign body may result in the counter-roller **48** being raised away from the first tool roller **34** in the direction **60**, in particular in opposition to the direction of gravity *g*. Disruptions such as deformed products, doubled products, skewed material webs, etc. can also result in raising away.

(86) In the event of such a movement, the support mode at the wedge device **62** is (briefly) canceled. As a result of the second roller **42** being moved away from the first roller **30** in the direction **60**, along the first axis of displacement **50**, the second wedge element **70** is detached from the first wedge element **68**—that is to say the second support face **76** is no longer in contact with the first support face **74**.

(87) The result is that the counter-force that has hitherto prevented the first wedge element **68** from being displaced in the direction **82**, in the support mode, is reduced such that, driven by the drive device **86**, the first wedge element **68** moves in the direction **82**.

(88) This is shown in FIG. 4(a). There, the wedge element **68** is displaced in the direction **82** by comparison with the position in FIG. 3(b), which shows the position of the first wedge element **68** during the cutting operation.

(89) In this case, the first wedge element **68** is displaced such that the first support face **74** abuts against the second support face **76** again.

(90) This displacement of the first wedge element **68** is performed automatically by the bias of the drive device **86** as soon as the counter-force on the wedge device **62** is reduced as a result of the

counter-roller **48** being raised away from the tool roller **34**.

(91) As a result of this movement of the first wedge element **68** in the direction **82**, the support mode is once again set, in which the second roller **42** (the counter-roller **48**) is supported on the machine stand **12** by way of the wedge device **62**.

(92) However, there is a spacing **102** (cf. FIG. **4(b)**, which corresponds to FIG. **4(a)**) between the second roller **42** and the tool roller **30**. This spacing is set automatically by the automatic displacement of the first wedge element **68**.

(93) As a result of the spacing **102**, the counter-roller **48** and the tool roller **34** are no longer in contact.

(94) If, starting from a support mode, the second roller **42** is raised away from the first roller **30**, then the corresponding first wedge element **68** follows automatically as a result of the drive device **86**, and a support mode is once again set, but in this case the second roller **42** is at a spacing from the first roller **30**. This prevents the second roller **42**, as a counter-movement after the raising away, from being displaced in the direction of the first roller **30** again on the axis of displacement **50** (that is to say, in the direction **58**) and in so doing prevents the tool roller **34** and the counter-roller **48** from impacting against one another. A mutual impact of this kind can result in damage to the blade **40** and/or the counter-roller **48**.

(95) To a certain extent, the automatic displacement of the first wedge element **68** in the direction **82** “freezes” raising away of the second roller **42** from the first roller **30** in a manner caused by disruptions in the material web **100**, in order to prevent the counter-roller **48** from impacting against the tool roller **34**.

(96) The sensor device **86** makes it possible to detect a disruption of this kind, which has resulted in a corresponding displacement of the first wedge element **68**. The displacement position of the first wedge element **68** can be used to detect the fact that the second roller **42** has been raised away from the first roller **30**.

(97) At that point, a proper cutting operation is no longer possible. By way of the evaluation device **98**, in particular the rotary cutting device **10** is then halted—that is to say that rotation of the first roller **30** and the second roller **42** is halted.

(98) For example, it is then provided for the first wedge element **68** to be reset to its displacement position for a cutting operation, this resetting being initiated by a user.

(99) The rotary cutting device **10** is for example used for manufacturing personal care items or packaging items.

(100) When operation of the rotary cutting device **10** is inactive, in particular the first wedge element **68** is not urged under force by the drive device **86**.

(101) Above, the first wedge element **68** was described as displaceable and coupled to the drive device **86**.

(102) As an alternative, it is also possible for the second wedge element **70** to be displaceable and in this case in particular to be displaceable in the direction **84**.

(103) It is also possible for both the first wedge element **68** and the second wedge element to be displaceable in opposing directions in order to provide securing against impact. A drive device then takes a corresponding form.

(104) As a result of the solution according to the invention, it is possible to provide a protection against impact between the counter-roller **48** and the tool roller **34** that is automated and in particular prevents damage from occurring at the counter-roller **48** and/or the tool roller **34** with a low hardening depth. A protection against roller impact is provided that is integrated into the rotary cutting device **10** in a structurally simple manner.

(105) The first wedge element **68** and the second wedge element **70** have a self-locking effect that prevents the raised roller (in the exemplary embodiment, the second roller **42**) from falling back into place.

(106) In the exemplary embodiment described above, the tool roller **34** is a cutting roller. It is also

possible for the tool roller, which acts on a workpiece and during operation “modifies” it, to be for example a stamping roller, bladed roller, squeezing roller or compactor roller, etc.

(107) A further exemplary embodiment of a rotary cutting device according to the invention, which is shown in FIG. 5 and designated **110**, comprises a machine stand **112** on which there is seated, by way of a bearing housing, as the first roller **114** a tool roller, for example a cutting roller. Further, a second roller **116** that is a counter-roller is seated on the machine frame **112** below the first roller **114**, in relation to the direction of gravity g .

(108) During a normal cutting operation, a third roller **118** acts on the first roller **114** and is positioned on the machine frame **112** above the first roller **114**, in relation to the direction of gravity g .

(109) The third roller **118** is a support roller that is supported against the first roller **114** and prevents the first roller **114** from undergoing excessive deflection during a cutting operation.

(110) The second roller **116** is held against the machine stand **112** such that it is displaceable on a first axis of displacement **50**.

(111) The third roller **118** is likewise movable such that it is displaceable on the first axis of displacement **50**.

(112) The second roller **116** can be pressed against the first roller **114** by way of a cutting pressure device **120**, for the purpose of setting a cutting force.

(113) A wedge device **122** having at least one displaceable wedge element is provided, and enables support of the second roller **116**.

(114) In principle, the wedge device **122** operates in the same way as the wedge device **62** described above.

(115) It is also possible in principle for the counter-roller to be arranged above the tool roller in relation to the direction of gravity g , and for the third roller (support roller) to be arranged below the tool roller in relation to the direction of gravity g .

(116) It is also possible to provide support for the counter-roller by way of a further roller.

(117) As a result of the wedge device **122**, a protection against roller impact can be achieved, as described above with reference to the rotary cutting device **10**; the possibility that after the second roller **116** has been raised away it can impact against the first roller **114** as a result of a disruption on a material web is prevented.

(118) A further exemplary embodiment of a rotary cutting device according to the invention, shown schematically in FIG. 6 and designated **10'**, takes basically the same form as the rotary cutting device **10**. Like reference numerals are used for like elements.

(119) The rotary cutting device **10'** comprises a sensor device **130** that serves to detect the position of the second roller **42** in relation to the machine stand **12**. The sensor device **130** is a distance-sensor device or position-sensor device that detects the position of the second roller **42** in relation to the machine stand **12** and hence also in relation to the first roller **30**.

(120) Sensor signals of the sensor device **130** are emitted to the evaluation device **98**. Further, for the purpose of controlling the drive device **86**, the evaluation device **98** is connected to the first drive **88** and the second drive **90** in a manner configured to transfer signals; the evaluation device **98** forms a controller for the drive device **86**.

(121) The sensor device **130** is connected to the evaluation device **98** in a manner configured to transfer signals. On the basis of sensor results from the sensor device **130**, the evaluation device **98** controls the drive device **86** having the first drive **88** and the second drive **90**.

(122) In one embodiment, the sensor device **130** comprises a first sensor **132** that is associated with the first frame element **22**. It further comprises a second sensor **134** that is associated with the second frame element **24**.

(123) In one embodiment, the first sensor **132** and the second sensor **134** are connected to the second roller **42** in a manner preventing displacement.

(124) Displacement of the second roller **42** brings about displacement of the sensors **132** and **134**.

(125) Arranged fixed to the first frame element **22** is a first encoder **136** that corresponds with the first sensor **132**. By way of the first sensor **132**, a position in relation to the first encoder **136** can be detected, or a spacing between the first sensor **132** and the first encoder **136** can be measured.

(126) Correspondingly, a second encoder **138** is arranged on the second frame element **24** in a manner preventing displacement, and cooperates with the second sensor **134**.

(127) In principle, it is also possible for the encoders to be connected to the second roller **42** in a manner preventing displacement and for the corresponding sensors of the sensor device **130** to be arranged fixed to the machine stand **12** (at the frame elements **22**, **24**). Furthermore, it is possible for one encoder to be seated on the machine stand **12** in a manner preventing displacement and for the other encoder to be prevented from displacement in relation to the second roller **42**, and correspondingly for the sensor associated with the encoder that is displaceable with the second roller **42** to be seated on the machine stand **12** in a manner preventing displacement; the sensor associated with the encoder that is fixed to the machine stand **12** is in that case displaceable with the second roller **42**.

(128) As mentioned, using the sensor device **130**, the position of the second roller **42** in relation to the machine stand **12** and hence also in relation to the first roller **30** and the wedge device **62** can be detected. In that case, using the sensor device **130**, the fact that the second roller **42** has been raised away and thus that a support of the second roller **42** against the wedge device **62** has been canceled can be detected. In particular, there is used as the detection result for raising away the fact that a particular threshold value determined by the sensor device **130** has been exceeded.

(129) The evaluation device **98** controls the drive device **86** in dependence on the sensor results from the sensor device **130**. In particular, if the particular threshold value is exceeded, the drive device **86** is operated such that the first wedge element **68** is displaced in such a way that support of the second roller **42** against the wedge device **62** is re-established, in which case the second roller **42** is at a spacing from the first roller **30**.

(130) As a result of the sensor device **130** and in cooperation with the evaluation device **98**, the drive device **86** can take a form such that there is no need for bias in relation to the first wedge element **68**. In that case, the drive device **86** can be controlled such that a force is only exerted on the first wedge element **68** once the above-mentioned threshold has been detected by the sensor device **130**. As a result, there is no need to compensate for example changes in the position of the first wedge element **68** in the support mode resulting from guidance play and/or vibrations at the rotary cutting device **10**.

(131) Using the sensor device **130** having the sensors **132**, **134**, it is possible to detect the position of the second roller **42** in the machine stand **12** both in relation to the first frame element **22** and the second frame element **24**. In particular, the first drive **88** and the second drive **90** are controlled separately. This allows optimized protection against roller impact to be achieved.

(132) In a further embodiment **110'** of a rotary cutting device according to the invention, which is shown schematically in FIG. 7 and takes basically the same form as the rotary cutting device **110**, there is provided a sensor device **150** that detects a position of the second roller **116** in relation to the machine stand **112**. The sensor device **150** has in particular a first sensor **152** and a second sensor **154** that are associated with opposing sides of the machine stand **12** (corresponding to the frame elements).

(133) In one embodiment, the first sensor **152** and the second sensor **154** are each fixedly connected to the machine stand **112**. Connected to the second roller **116** in a manner preventing displacement are a first encoder **156** and a second encoder **158**. The first sensor **152** cooperates with the first encoder **156**; the second sensor **154** cooperates with the second encoder **158**. Using the sensor device **150**, it is possible to detect the spacing between the first sensor **152** and the first encoder **156**, and between the second sensor **154** and the second encoder **158**. In this way, the position of the second roller **116** on the machine stand **112** and hence the position of the second roller **116** in relation to the wedge device **122** and that of the second roller **116** in relation to the

first roller **114** can be detected.

(134) The sensor device **150** is connected to the corresponding evaluation device **98** in a manner configured to transfer signals, and delivers its sensor signals thereto. The evaluation device **98** controls a drive device **160** having a first drive **162** and a second drive **164** for the wedge device **122**. The evaluation device **98** is a controller for the drive device **160**.

(135) As described above with reference to the rotary cutting device **10'**, if a particular threshold value of the position of the second roller **116** on the machine stand **112** is detected, the drive device **160** is controlled such that the corresponding first wedge element of the wedge device **122** is displaced in such a way that once support is canceled by displacing the displaceable wedge element support is automatically re-established.

LIST OF REFERENCE NUMERALS

(136) **10, 10'** Rotary cutting device **12** Machine stand **14** Base **16** Frame **18** Bearer **20** Transverse strut **22** First frame element **24** Second frame element **26** Strut **28** First bearing housing **30** First roller **32** Axis of rotation **34** Tool roller **36** First support ring **38** Second support ring **40** Blade **42** Second roller **44** Second bearing housing **46** Axis of rotation **48** Counter-roller **50** First axis of displacement **52** Cutting pressure device **54** First sub-part **56** Second sub-part **58** Direction **60** Opposing direction **62** Wedge device **64** First part-device **66** Second part-device **68** First wedge element **70** Second wedge element **72** Second axis of displacement **74** First support face **76** Second support face **78** Lower side **80** Upper side **82** Direction **84** Direction **86** Drive device **88** First drive **90** Second drive **91** Sensor device **91a** Connection configured to transfer signals **91b** Connection configured to transfer signals **92** Pneumatic cylinder **94** Connection **96** Sensor device **98** Evaluation device **100** Material web **102** Spacing **110, 110'** Rotary cutting device **112** Machine stand **114** First roller **116** Second roller **118** Third roller **120** Cutting pressure device **122** Wedge device **130** Sensor device **132** First sensor **134** Second sensor **136** First encoder **138** Second encoder **150** Sensor device **152** First sensor **154** Second sensor **156** First encoder **158** Second encoder **160** Drive device **162** First drive **164** Second drive

Claims

1. A rotary cutting device, comprising: a machine stand; a first roller that is mounted rotatably on the machine stand; a second roller that is mounted rotatably on the machine stand; wherein either (i) the first roller is a tool roller and the second roller is a counter-roller, or (ii) the second roller is a tool roller and the first roller is a counter-roller; and wherein the second roller is mounted on the machine stand such that the second roller is displaceable on a first displacement axis; and a cutting pressure device by way of which a cutting pressure is configured to be exerted between the second roller and the first roller; wherein the second roller is displaceable during a cutting operation; wherein, in a support mode, the second roller is supported against a wedge device; wherein the wedge device has at least one displaceable wedge element; wherein a displacement position of the at least one displaceable wedge element predetermines an initial spacing between the second roller and the first roller; wherein there is associated with the at least one displaceable wedge element a drive device for a displacement movement of the at least one displaceable wedge element on a second displacement axis; wherein the drive device is adapted such that, when the second roller is raised relatively away from the first roller during the cutting operation and the second roller is no longer supported against the wedge device, the at least one displaceable wedge element is displaced automatically such that the second roller is once again supported against the wedge device, in which case the second roller is at an increased fixed spacing from the first roller which is larger than the initial spacing; and wherein the drive device is adapted such that the drive device constantly exerts a biasing force on the at least one displaceable wedge element during the cutting operation.

2. The rotary cutting device as claimed in claim 1, wherein, during the cutting operation, the first

roller is supported against the second roller by way of at least one support ring, which is arranged on at least one of the first roller and the second roller.

3. The rotary cutting device as claimed in claim 2, wherein: the at least one support ring comprises a first support ring and a second support ring that are arranged on the tool roller, and a blade is positioned on the tool roller between the first support ring and the second support ring in relation to a direction parallel to an axis of rotation of the tool roller.

4. The rotary cutting device as claimed in claim 1, wherein the drive device is adapted such that a movement of the at least one displaceable wedge element is driven in a direction in which the initial spacing between the second roller and the first roller is increased to the increased spacing if the second roller is supported against the wedge device.

5. The rotary cutting device as claimed in claim 1, wherein the biasing force is subject to closed-loop control.

6. The rotary cutting device as claimed in claim 1, having a sensor device that detects a position of the second roller in relation to the machine stand.

7. The rotary cutting device as claimed in claim 6, wherein the sensor device is connected, in a manner configured to transfer signals, to an evaluation device, and the evaluation device controls the drive device in dependence on signals from the sensor device.

8. The rotary cutting device as claimed in claim 7, wherein the evaluation device controls the drive device such that, when the sensor device detects a threshold value that corresponds to the second roller being raised away from the first roller and to the second roller no longer being supported against the wedge device, the at least one displaceable wedge element is displaced.

9. The rotary cutting device as claimed in claim 1, wherein the support mode prevails during a normal cutting operation, and prevails if, after the second roller has been raised relatively away from the first roller, the at least one displaceable wedge element has been displaced for the purpose of the second roller being once again supported against the wedge device.

10. The rotary cutting device as claimed in claim 1, wherein the wedge device comprises at least one part-device having a first wedge element and a second wedge element, wherein in the support mode the first wedge element is supported against the second wedge element, and wherein at least one of the first wedge element and the second wedge element is displaceable and coupled to the drive device.

11. The rotary cutting device as claimed in claim 10, wherein the first wedge element is associated with the first roller and the second wedge element is associated with the second roller.

12. The rotary cutting device as claimed in claim 10, wherein the first wedge element is connected to the first roller in a manner preventing displacement relative to the first axis of displacement.

13. The rotary cutting device as claimed in claim 10, wherein the first wedge element is arranged on the machine stand in a manner preventing displacement relative to the first axis of displacement.

14. The rotary cutting device as claimed in claim 10, wherein the second wedge element is connected to the second roller in a manner preventing displacement relative to the first axis of displacement.

15. The rotary cutting device as claimed in claim 14, wherein the second wedge element is displaceable with the second roller on the first axis of displacement.

16. The rotary cutting device as claimed in claim 10, wherein the first wedge element is arranged on a first bearing housing of the first roller, by way of which the first roller is seated on the machine stand.

17. The rotary cutting device as claimed in claim 10, wherein the second wedge element is arranged on a second bearing housing by way of which the second roller is seated on the machine stand.

18. The rotary cutting device as claimed in claim 1, wherein the first roller is seated on the machine stand in a manner preventing displacement in relation to the first axis of displacement.

19. The rotary cutting device as claimed in claim 1, wherein the first axis of displacement is

oriented transversely to the second axis of displacement.

20. The rotary cutting device as claimed in claim 1, wherein at least one of the first axis of displacement and the second axis of displacement are oriented transversely to an axis of rotation of the first roller.

21. The rotary cutting device as claimed in claim 1, wherein the wedge device has a first part-device and a second part-device, wherein the first part-device and the second part-device are at a spacing from one another in a direction parallel to an axis of rotation of the first roller, and in the support mode the second roller is supported both against the first part-device and against the second part-device.

22. The rotary cutting device as claimed in claim 21, wherein the first part-device and the second part-device take the same form.

23. The rotary cutting device as claimed in claim 21, wherein the drive device has a first drive for the first part-device and a second drive for the second part-device, wherein the first drive and the second drive are synchronized.

24. The rotary cutting device as claimed in claim 1, wherein the drive device is or comprises a mechanical drive or pneumatic drive or hydraulic drive or magnetic drive or inductive drive or electromagnetic drive or motorized drive.

25. The rotary cutting device as claimed in claim 1, wherein the tool roller is a cutting roller or stamping roller or bladed roller or compactor roller or squeezing roller.

26. The rotary cutting device as claimed in claim 1, wherein at least one of the first roller and the second roller are supported by at least one further roller.

27. The rotary cutting device as claimed in claim 1, wherein, during a cutting operation, displaceability of the at least one displaceable wedge element is released and the drive device acts on the at least one displaceable wedge element, and wherein a cutting force is set between the first roller and the second roller by way of the cutting pressure device, wherein the wedge device and the drive device take a form adapted to one another such that during disruption-free operation the second roller is pressed against the first roller, and the initial spacing between the first roller and the second roller is fixed.

28. The rotary cutting device as claimed in claim 27, wherein, when the second roller is raised away from the first roller because of a disruption in a material web, the drive device displaces the at least one displaceable wedge element such that, because support of the second roller against the wedge device is then re-established, the second roller cannot impact against the first roller.

29. The rotary cutting device as claimed in claim 1, having a sensor device that determines a displacement position of the at least one displaceable wedge element and determines the displacement position of the at least one displaceable wedge element in relation to the machine stand.

30. The rotary cutting device as claimed in claim 29, having at least one of a display device and an evaluation device that is coupled to the sensor device in a manner configured to transfer signals.

31. The rotary cutting device as claimed in claim 1, wherein in the event that the at least one displaceable wedge element is automatically displaced, the cutting operation is stopped until the initial spacing between the first roller and the second roller can be reset.

32. A method for operating a rotary cutting device, in which, during a machining operation, a first roller is supported against a second roller, and a material web is guided between the first roller and the second roller, wherein: (i) the first roller is a tool roller and the second roller is a counter-roller, or (ii) the second roller is a tool roller and the first roller is a counter-roller, and wherein the second roller is displaceable in relation to the first roller on a first displacement axis, in a support mode, the second roller is supported against a wedge device, at least one movable wedge element is displaced such that, when the second roller is raised away from the first roller during the machining operation, the second roller is once again supported against the wedge device, the second roller is then positioned at an initial spacing from the first roller, the at least one displaceable wedge

element is displaced automatically such that, when the second roller has been raised away from the first roller during the machining operation, the second roller is prevented from impacting against the first roller because the second roller is supported against the wedge device at an increased fixed spacing which is larger than the initial spacing, and a drive device constantly exerts a biasing force on the at least one displaceable wedge element during the machining operation.

33. The method as claimed in claim 32, wherein the displacement movement of the at least one displaceable wedge element is performed automatically.

34. The method as claimed in claim 32, wherein, during the machining operation, the first roller and the second roller are operated at the same speed of revolution.

35. The method as claimed in claim 32, further comprising: stopping the machining operation when the at least one displaceable wedge element is automatically displaced; and resetting the initial spacing between the first roller and the second roller before resuming the machining operation.
